

Machine Project II

CS 521 — Machine Learning and Compilers
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Student Name: **Zirui Zhou** NetID: **ziruiz8**

1 GPU

1.1 Part I

1. Suppose we use A_{ij} to denote the element in matrix A . For reduction we have

$$O_j = \sum_i A_{ij},$$

so the einsum notation could be $ij \rightarrow j$.

2. For convolution we have

$$O_{ij} = \sum_{k,l} A_{i+k,j+l} B_{kl},$$

so the einsum notation could be $ijkl, kl \rightarrow ij$. Since standard einsum cannot express arithmetic on indices, there is no way to express the convolution operation in einsum notation.

3. For batch matrix multiplication we have

$$O_{ijl} = \sum_k A_{ijk} B_{ikl},$$

so the einsum notation could be $ijk, ikl \rightarrow ijl$.

4. For `Relu` we have

$$O_{ij} = \max(A_{ij}, 0),$$

so the einsum notation could be $ij \rightarrow ij$. Since Einsum cannot apply arbitrary nonlinear functions such as $\max(x, 0)$, there is no way to express the `Relu` operation in einsum notation.

5. For the elementwise multiplication we have

$$O_{ijk} = A_{ijk} B_{ijk},$$

so the einsum notation could be $ijk, ijk \rightarrow ijk$.

1.2 Part II

1. The TCG of original computation is shown in Figure 1a.
2. After applying $R1$ then $R2$, the TCG is shown in Figure 1b.
3. After applying $R2$ then $R1$, the TCG is shown in Figure 1c. Note that $R1$ is not applicable after $R2$ here.
4. The difference is that in Figure 1c there is still `01`, while in Figure 1b `01` is substituted by a new matrix multiplication. This is because in Figure 1b we first apply $R1$ that could substitute `01` with a new matrix multiplication, while in Figure 1c we first apply $R2$ and it breaks the pattern that $R1$ could be applied to `01`.

5. First do the shape analysis. We have

$$A : [N, M], B : [M, K], C : [U, N], D : [N, V].$$

For Figure 1b, the addition and multiplication times is $\Theta(UN(M+V) + UMK)$. For Figure 1c, the addition and multiplication times is $\Theta(NMK + UN(K+V))$. If Figure 1b is better than Figure 1c, we have

$$UM(N+K) < NK(U+M).$$

There are many possible values of U, M, N, K that could satisfy this inequality. For example, when U is extremely smaller than M, N and K , we have $UM(N+K) \approx M(N+K) < NMK$ and $NK(U+M) \approx NMK$.

Note that the profitability condition does not relate to V . This is because rule $R2$ does not reduce the computation strength, so the profitability condition is only related to the computation strength change by applying $R1$.

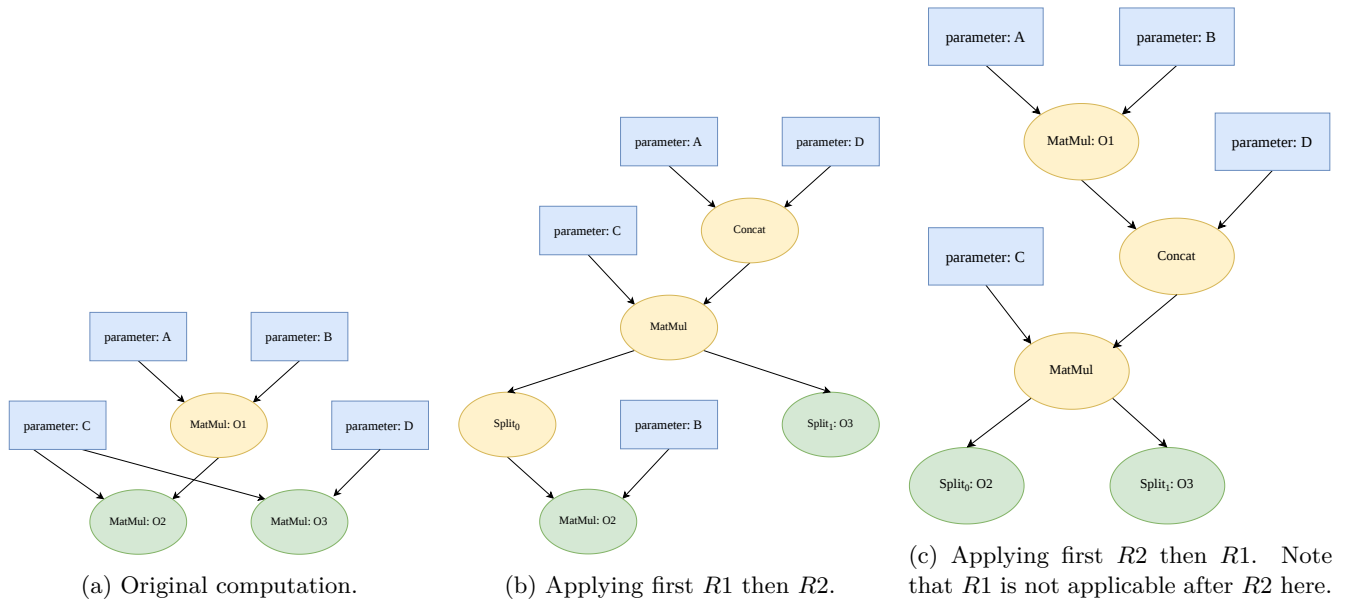


Figure 1: TCG of Computation.

1.3 Part III